

SpaceX Falcon 9 First Stage Landing Prediction

Data Analysis and Machine Learning for Predicting First-Stage Landing Success

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IBM Data Science Capstone Project

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Executive Summary

- SpaceX's ability to reuse Falcon 9 first stages significantly reduces launch costs. This project analyzed historical Falcon 9 launch data to identify the factors associated with successful first-stage landings and developed machine learning classification models to predict landing outcomes.
- The analysis combined data collected from the SpaceX API and Wikipedia web scraping, followed by data wrangling, exploratory data analysis, SQL analysis, interactive mapping, and dashboard visualization.
- Multiple classification algorithms were evaluated using GridSearchCV, with the best-performing model selected based on test accuracy.

Introduction

- **The Business Problem**

SpaceX advertises Falcon 9 launches at a significantly lower price than many competitors, partly because of first-stage reusability.

The key question was:

Can we predict whether the Falcon 9 first stage will successfully land based on information available before launch?

- **Project Objectives**

1. Collect Falcon 9 launch data.
2. Clean and prepare the data for analysis.
3. Explore relationships between launch characteristics and landing success.
4. Perform SQL-based analysis.
5. Visualize launch locations geographically.
6. Build an interactive dashboard.
7. Develop machine-learning models to predict landing success.

DATA COLLECTION & WRANGLING

1. SpaceX API

- The API-based workflow collected information such as:
- Flight number
- Launch date
- Booster version
- Payload mass
- Launch site
- Orbit
- Landing outcome

2. Web Scraping

- Historical launch information was extracted from Wikipedia tables using web scraping techniques.

Data Wrangling Methodology

Key activities included:

- Identifying missing values.
- Calculating missing payload values where necessary.
- Removing or handling incomplete records.
- Standardizing categorical variables.
- Creating the binary Class target variable.
- Preparing the dataset for exploratory and predictive analysis.

EDA Methodology

Exploratory Data Analysis

The dataset was investigated using visualizations to understand relationships between:

- Flight number and launch success.
- Payload mass and launch success.
- Launch site and landing success.
- Orbit type and landing success.
- Booster version and landing success.

Interactive Visual Analytics Methodology

Tools Used

Folium

- Interactive launch-site maps.
- Marker clusters.
- Launch proximity analysis.

Plotly Dash

- Interactive launch-site dropdown.
- Success/failure pie charts.
- Payload range selection.
- Scatter plots.

EDA with SQL Results

Tasks

Now write and execute SQL queries to solve the assignment tasks.

Note: If the column names are in mixed case enclose it in double quotes For Example "Landing_Outcome"

Task 1

Display the names of the unique launch sites in the space mission

```
[10]: %%sql
      SELECT DISTINCT "Launch_Site"
      FROM SPACEXTABLE;

* sqlite:///my_data1.dbDone.
```

```
[10]: Launch_Site
      CCAFS LC-40
      VAFB SLC-4E
      KSC LC-39A
      CCAFS SLC-40
```


Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[11]: %%sql
SELECT *
FROM SPACEXTABLE
WHERE "Launch_Site" LIKE 'CCA%'
LIMIT 5;
```

```
* sqlite:///my_data1.db
Done.
```

```
[11]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[12]: %%sql
      SELECT SUM("Payload_Mass__kg_") AS total_payload_mass
      FROM SPACEXTABLE
      WHERE "Customer" = 'NASA (CRS)';

      * sqlite:///my_data1.db
      Done.
```

```
[12]: total_payload_mass
      45596
```

 Task 4



Display average payload mass carried by booster version F9 v1.1

```
[13]: %%sql
      SELECT AVG("Payload_Mass__kg_") AS average_payload_mass
      FROM SPACEXTABLE
      WHERE "Booster_Version" = 'F9 v1.1';

      * sqlite:///my_data1.db
      Done.
```

```
[13]: average_payload_mass
      2928.4
```

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[14]: %%sql
      SELECT MIN("Date") AS first_successful_ground_pad_landing
      FROM SPACEXTABLE
      WHERE "Landing_Outcome" = 'Success (ground pad)';

      * sqlite:///my_data1.db
      Done.
```

```
[14]: first_successful_ground_pad_landing
      2015-12-22
```

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[15]: %%sql

      SELECT "Booster_Version"
      FROM SPACEXTABLE
      WHERE "Landing_Outcome" = 'Success (drone ship)'
      AND "Payload_Mass_kg_" > 4000
      AND "Payload_Mass_kg_" < 6000;

      * sqlite:///my_data1.db
```

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

[15]: %%sql

```
SELECT "Booster_Version"  
FROM SPACEXTABLE  
WHERE "Landing_Outcome" = 'Success (drone ship)'  
      AND "Payload_Mass__kg_" > 4000  
      AND "Payload_Mass__kg_" < 6000;
```

* sqlite:///my_data1.db

Done.

[15]: **Booster_Version**

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Task 7

List the total number of successful and failure mission outcomes

```
[16]: %%sql

SELECT
    CASE
        WHEN "Landing_Outcome" LIKE 'Success%' THEN 'Success'
        ELSE 'Failure'
    END AS mission_outcome,
    COUNT(*) AS total
FROM SPACEXTABLE
GROUP BY mission_outcome;
```

```
* sqlite:///my_data1.db
```

Done.

```
[16]: mission_outcome  total
```

Failure	1
Failure	98
Failure	1
Success	1

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

[17]: %%sql

```
SELECT DISTINCT "Booster_Version"  
FROM SPACEXTABLE  
WHERE "Payload_Mass__kg_" = (  
    SELECT MAX("Payload_Mass__kg_")  
    FROM SPACEXTABLE  
);
```

* sqlite:///my_data1.db
Done.

[17]: **Booster_Version**

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
[18]: %%sql

SELECT substr("Date", 6, 2) AS Month,
       "Landing_Outcome",
       "Booster_Version",
       "Launch_Site"
FROM SPACEXTABLE
WHERE substr("Date", 0, 5) = '2015'
   AND "Landing_Outcome" = 'Failure (drone ship)';

* sqlite:///my_data1.db
Done.
```

```
[18]:
```

	Month	Landing_Outcome	Booster_Version	Launch_Site
	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

▼ Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

[19]:

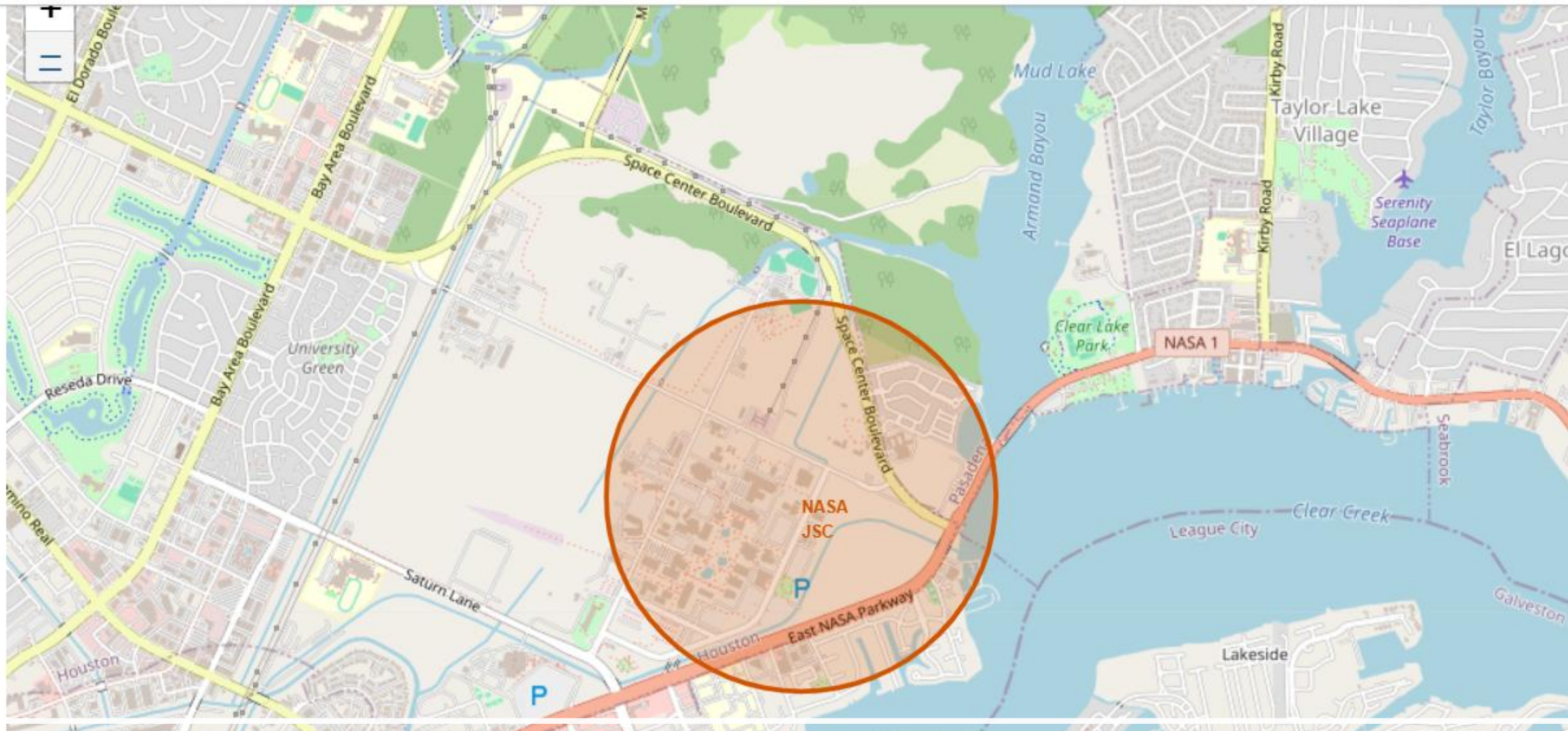
```
%%sql

SELECT "Landing_Outcome",
       COUNT(*) AS "Count"
FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY "Count" DESC;
```

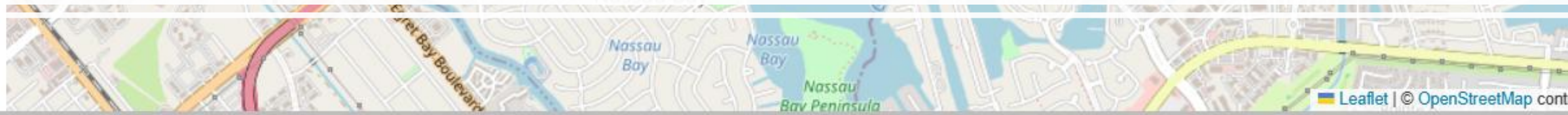
```
* sqlite:///my_data1.db
Done.
```

[19]:

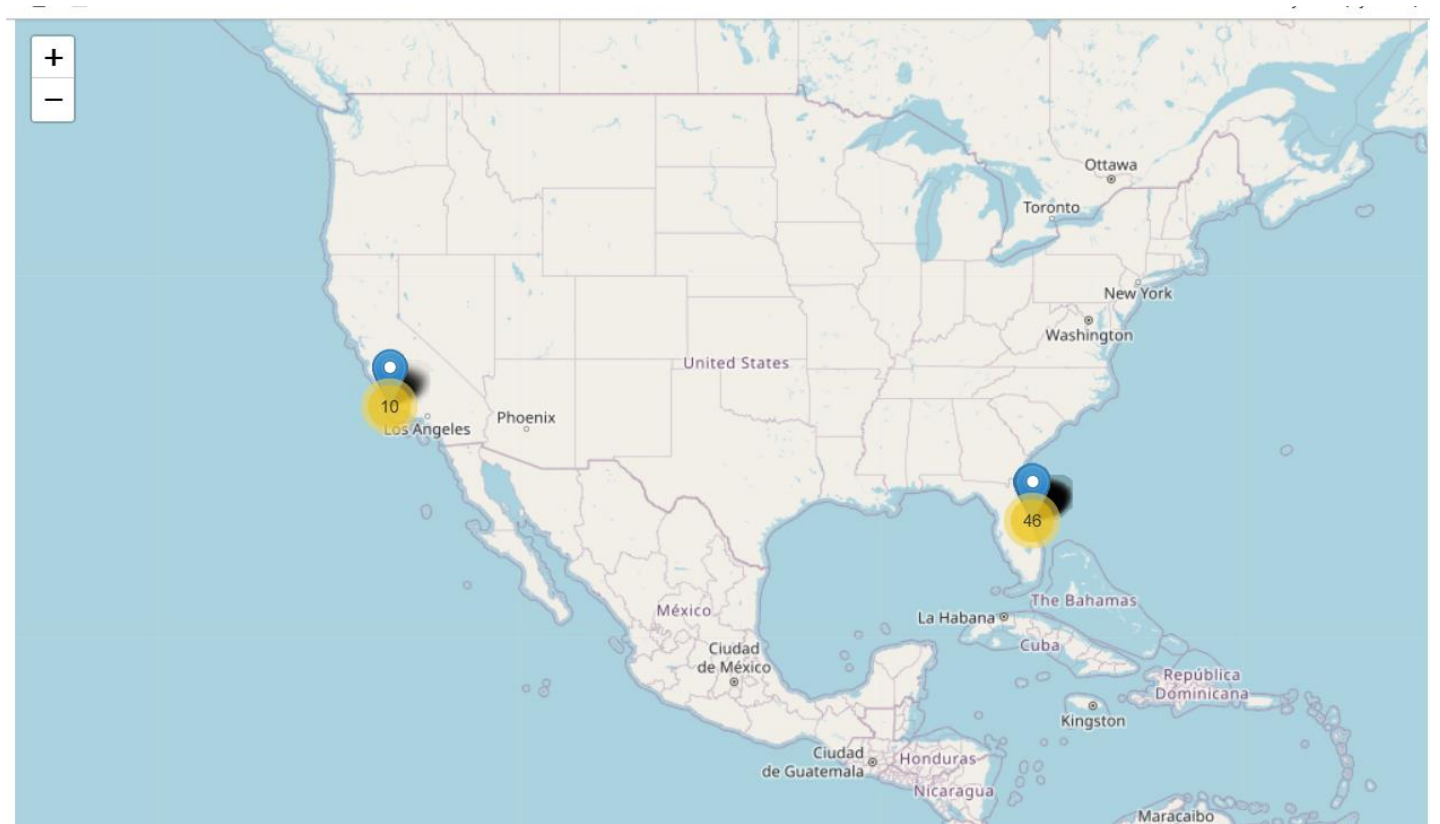
Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
-	-

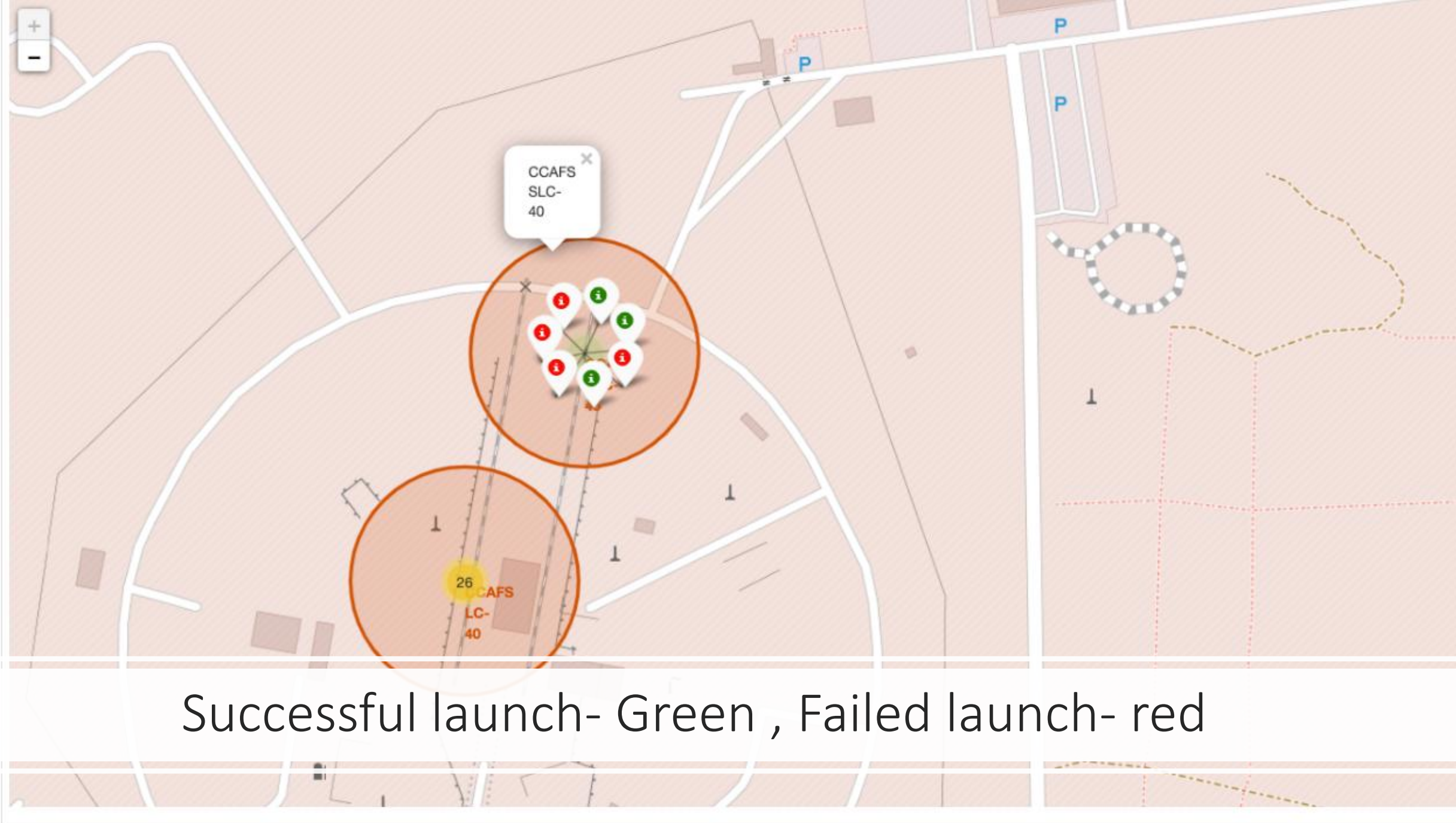


FOLIUM MAP NSSA SPACE CENTER



LAUNCH SITES





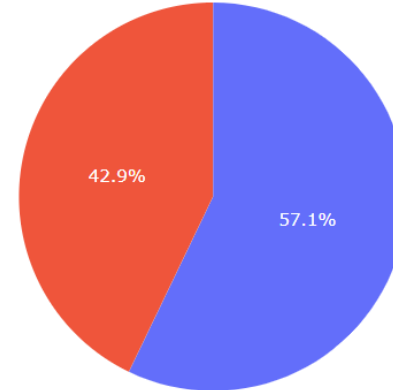
PLOTLY DASHBOARD – LAUNCH SUCCESS

SpaceX Launch Records Dashboard

All Sites

✕ ▾

Total Launch Success



0
1

0



0

1000

2000

3000

4000

5000

6000

7000

8000

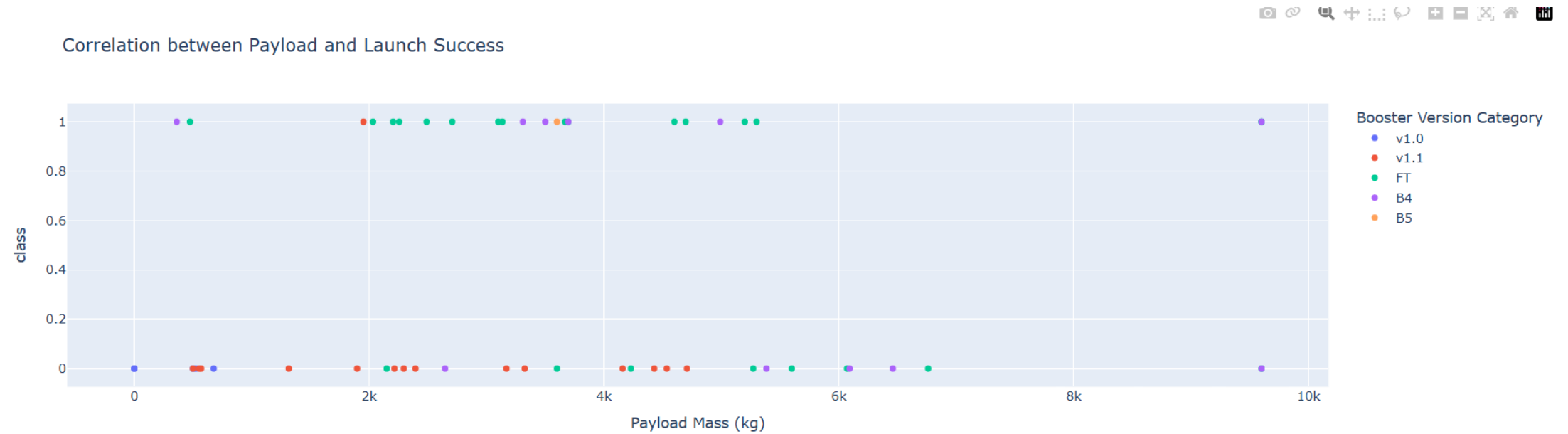
9000

10k



9600

Correlation between Payload and Launch Success



Predictive Analysis Methodology

Creating a machine-learning pipeline:

- **Prepared Dataset**
- ↓
- **Feature Selection**
- ↓
- **One-Hot Encoding**
- ↓
- **Train/Test Split**
- ↓
- **Feature Scaling**
- ↓
- **Model Training**
- ↓
- **GridSearchCV Hyperparameter Tuning**
- ↓
- **Model Evaluation**

Models included in the project:

- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors

Hyperparameters were optimized using GridSearchCV with cross-validation to identify the best configuration for each classifier.

Predictive Analysis Results

Model Comparison Summary

- After hyperparameter tuning with GridSearchCV (cv=10):

Test Accuracy Results:

- **Logistic Regression:** 83.33%
- **Support Vector Machine (SVM):** 83.33%
- **Decision Tree Classifier:** 77.78%
- **K-Nearest Neighbors (KNN):** 83.33%

Conclusion

- Three models (Logistic Regression, SVM, and KNN) achieved the highest test accuracy of 83.33%. The SVM achieved the highest validation accuracy (84.82%), making it a strong choice for this prediction task.